

MARKET ANALYSIS

AI Data Centers' 125 GW Gap Forces Utilities and Developers Into a New Capital Calculus

BofA sees a 100 GW supply gap by 2030, pushing developers toward onsite gas and batteries as grid capacity lags.

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A source-grounded Light Tower Insight. The complete note follows.

The headline number from Bank of America is arresting: data centers could add 125 GW of U.S. electric load by 2030, while regulated utilities are on track to bring only 93 GW of new accredited supply online. That leaves a gap of more than 100 GW. The conventional reading is that the grid is unprepared for AI-driven demand. That is true but incomplete. The more revealing story is about capital allocation under constraint.

The gap is not a forecast of blackouts. It is a forecast of where capital will flow next. When the grid cannot deliver, developers do not stop building. They build their own power. BofA reports that more than 7.5 GW of data center projects with onsite generation are already under construction, with another 60 GW-plus in pre-construction. Those numbers are not small. They represent a structural shift in how large-scale computing infrastructure gets financed and built.

The mechanism is straightforward. Data center developers need firm, reliable power to run AI chips that consume three to five times the electricity of conventional servers. Utilities cannot deliver it fast enough because new transmission takes years to permit and build, and large gas turbines are largely sold out through 2030. So developers are turning to behind-the-meter gas reciprocating engines and battery storage. These assets can be deployed in months, not years, and they give the developer control over the most critical input: uptime.

This changes the capital stack for data center projects. Onsite generation is not cheap. A gas engine plant can cost \$1,500 to \$2,000 per kilowatt, and batteries add another \$400 to \$600 per kilowatt-hour. For a 100 MW data center, that is \$150 million to \$200 million in additional capital that sits outside the traditional building cost. That capital must come from somewhere, and it competes with the server, cooling, and real estate spend.

The implication for lenders and equity investors is that data center underwriting now includes a power-generation component that was previously the utility's responsibility. That adds construction risk, fuel-price risk, and operational complexity. It also adds a new source of return. Developers who can finance and operate onsite generation effectively will have a competitive advantage in site selection and speed to market. Those who cannot will wait for grid upgrades that may not arrive before 2032.

The BofA analysts also note that utilities are delaying coal plant retirements across at least seven states to preserve dispatchable capacity. That is a signal that the regulatory compact is shifting. Regulators are allowing utilities to keep older, higher-emission assets online because the reliability cost of retiring them is now higher than the environmental cost of running them. That creates a tension for investors in renewable generation and battery storage, who had counted on coal retirements to open market share.

For commercial real estate owners and developers, the story is not about data centers alone. It is about what happens to industrial and warehouse sites near constrained grids. If utilities cannot serve new load, the value of land with existing high-capacity grid connections rises. Sites that can accommodate onsite generation or that sit near substations with available capacity become scarce assets. That is a basis play that institutional capital is already testing.

The open question is whether the 100 GW gap will be filled by onsite generation, transmission upgrades, or demand destruction. BofA's report suggests the market is betting on onsite generation, at least in the near term. But the economics of behind-the-meter power depend on natural gas prices, battery costs, and the willingness of data center operators to tie up capital in generation assets rather than computing hardware. That trade-off will define the next phase of AI infrastructure investment.

The market is not waiting for the grid to catch up. It is building its own solution, one gas engine at a time. The question for lenders and equity partners is whether they are underwriting the power plant as well as the data hall.

SOURCES

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<https://www.constructiondive.com/news/ai-data-center-growth-utilities-generation-plans/825631/>

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